

## Bootstrap 4

If you don't want to download and host Bootstrap 4 yourself, you can include it from a CDN (Content Delivery Network).

MaxCDN provides CDN support for Bootstrap's CSS and JavaScript. You must also include jQuery:

```
<!-- Latest compiled and minified CSS -->
<link rel="stylesheet" href="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.0/css/bootstrap.min.css">

<!-- jQuery library -->
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>

<!-- Popper JS -->
<script src="https://cdnjs.cloudflare.com/ajax/libs/popper.js/1.16.0/umd/popper.min.js"></script>

<!-- Latest compiled JavaScript -->
<script src="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.0/js/bootstrap.min.js"></script>
```

### Grid Options

列表如下。

|                 | Extra small devices<br>Phones (<768px)                                      | Small devices<br>Tablets (≥768px) | Medium devices<br>Desktops (≥992px) | Large devices<br>Desktops (≥1200px) |
|-----------------|-----------------------------------------------------------------------------|-----------------------------------|-------------------------------------|-------------------------------------|
| Grid behavior   | Horizontal at all times<br>Collapsed to start, horizontal above breakpoints |                                   |                                     |                                     |
| Container width | None (auto)                                                                 | 750px                             | 970px                               | 1170px                              |
| Class prefix    | .col-xs-                                                                    | .col-sm-                          | .col-md-                            | .col-lg-                            |
| # of columns    | 12                                                                          |                                   |                                     |                                     |
| Column width    | Auto                                                                        | ~62px                             | ~81px                               | ~97px                               |
| Gutter width    | 30px (15px on each side of a column)                                        |                                   |                                     |                                     |
| Nestable        | Yes                                                                         |                                   |                                     |                                     |
| Offsets         | Yes                                                                         |                                   |                                     |                                     |
| Column ordering | Yes                                                                         |                                   |                                     |                                     |

因此我們可以依此做基本的設定，在 Desktop 的結果如下：

|           |           |           |           |           |           |           |           |           |           |           |           |
|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| .col-md-1 | .col-md-1 | .col-md-1 | .col-md-1 | .col-md-1 | .col-md-1 | .col-md-1 | .col-md-1 | .col-md-1 | .col-md-1 | .col-md-1 | .col-md-1 |
| .col-md-8 |           |           |           |           |           |           |           | .col-md-4 |           |           |           |
| .col-md-4 |           |           |           | .col-md-4 |           |           |           | .col-md-4 |           |           |           |
| .col-md-6 |           |           |           |           |           | .col-md-6 |           |           |           |           |           |

## 規則

Grid System 是經由 Row（列）和 Column（行）來建立頁面的架構的，然後再將內容裝在這些由 Row（列）和 Column（行）組成的框框裡面。

簡述規則如下：

- class 的結構依序為：  
.container( 固定寬度)或 .container-fluid -> .row -> Column。 .container 或 .container-fluid 讓版面有適當的對齊方式（alignment）和間格（padding）。
- 使用水平方向的 .row 來群組 Column。
- 內容放在 Column 之內，且 Column 一定緊接在 .row 之下，是為 Immediate Children。
- 使用 class .row 或 .col-xs-4 來建立頁面的架構，也可以使用 Less mixins and variables 來做設定或調整。
- Column 為最小單位的方格，且有間格將彼此格開，並由 .row 使用負的 margin 值校正因 Column 而多出來的左右 padding。
- 指定 Column 的格數（最多到 12），例如一列希望有 3 個相等的 Column，可指定 3 個 .col-xs-4。
- 基本上一個 Row 放置 12 個 Column，若有一個 Row 超過 12 個 Column，則會斷行放置多出來的 Column。
- 使用 Grid Class 會影響到大於或等於設定分段點的 Device，例如：使用 .col-md-\* 不僅會影響到 Desktop，若沒有設定 .col-lg-\*, 還會影響到 Large Desktop。

## Media Queries 的分段點

- .col- (extra small devices - screen width less than 576px)
- .col-sm- (small devices - screen width equal to or greater than 576px)
- .col-md- (medium devices - screen width equal to or greater than 768px)
- .col-lg- (large devices - screen width equal to or greater than 992px)
- .col-xl- (xlarge devices - screen width equal to or greater than 1200px)

## Bootstrap 4 Table

| Firstname | Lastname | Email            |
|-----------|----------|------------------|
| John      | Doe      | john@example.com |
| Mary      | Moe      | mary@example.com |
| July      | Dooley   | july@example.com |

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.0/css/bootstrap.min.css">
  <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
  <script src="https://cdnjs.cloudflare.com/ajax/libs/popper.js/1.16.0/umd/popper.min.js"></script>
  <script src="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.0/js/bootstrap.min.js"></script>
</head>
<body>

<div class="container">
  <h2>Hover Rows</h2>
  <p>The .table-hover class enables a hover state (grey background on mouse over) on table
rows:</p>
  <table class="table table-hover">
    <thead>
      <tr>
        <th>Firstname</th>
        <th>Lastname</th>
        <th>Email</th>
      </tr>
    </thead>
    <tbody>
      <tr>
        <td>John</td>
        <td>Doe</td>
        <td>john@example.com</td>
```

```

        </tr>
        <tr>
            <td>Mary</td>
            <td>Moe</td>
            <td>mary@example.com</td>
        </tr>
        <tr>
            <td>July</td>
            <td>Dooley</td>
            <td>july@example.com</td>
        </tr>
    </tbody>
</table>
</div>
</body>
</html>

```

| Firstname | Lastname | Email            |
|-----------|----------|------------------|
| John      | Doe      | john@example.com |
| Mary      | Moe      | mary@example.com |
| July      | Dooley   | july@example.com |

```

<!DOCTYPE html>
<html lang="en">
<head>
    <title>Bootstrap Example</title>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.0/css/bootstrap.min.css">
    <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
    <script src="https://cdnjs.cloudflare.com/ajax/libs/popper.js/1.16.0/umd/popper.min.js"></script>
    <script src="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.0/js/bootstrap.min.js"></script>
</head>
<body>

```

```
<div class="container">
  <h2>Dark Striped Table</h2>
  <p>Combine .table-dark and .table-striped to create a dark, striped table:</p>
  <table class="table table-dark table-striped">
    <thead>
      <tr>
        <th>Firstname</th>
        <th>Lastname</th>
        <th>Email</th>
      </tr>
    </thead>
    <tbody>
      <tr>
        <td>John</td>
        <td>Doe</td>
        <td>john@example.com</td>
      </tr>
      <tr>
        <td>Mary</td>
        <td>Moe</td>
        <td>mary@example.com</td>
      </tr>
      <tr>
        <td>July</td>
        <td>Dooley</td>
        <td>july@example.com</td>
      </tr>
    </tbody>
  </table>
</div>
</body>
</html>
```

| Class            | Description                                                  |
|------------------|--------------------------------------------------------------|
| .table-primary   | Blue: Indicates an important action                          |
| .table-success   | Green: Indicates a successful or positive action             |
| .table-danger    | Red: Indicates a dangerous or potentially negative action    |
| .table-info      | Light blue: Indicates a neutral informative change or action |
| .table-warning   | Orange: Indicates a warning that might need attention        |
| .table-active    | Grey: Applies the hover color to the table row or table cell |
| .table-secondary | Grey: Indicates a slightly less important action             |
| .table-light     | Light grey table or table row background                     |
| .table-dark      | Dark grey table or table row background                      |

...

```
<tr class="table-success">
  <td>Success</td>
  <td>Doe</td>
  <td>john@example.com</td>
</tr>
```

...

.table-responsive class adds a scrollbar to the table when needed (when it is too big horizontally):

```
<div class="table-responsive">
  <table class="table">
    ...
  </table>
</div>
```

## Bootstrap 4 Image Shapes



```



```

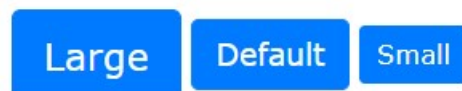
## Button Style



```
<button type="button" class="btn">Basic</button>
<button type="button" class="btn btn-primary">Primary</button>
<button type="button" class="btn btn-secondary">Secondary</button>
<button type="button" class="btn btn-success">Success</button>
<button type="button" class="btn btn-info">Info</button>
<button type="button" class="btn btn-warning">Warning</button>
<button type="button" class="btn btn-danger">Danger</button>
<button type="button" class="btn btn-dark">Dark</button>
<button type="button" class="btn btn-light">Light</button>
<button type="button" class="btn btn-link">Link</button>
```



```
<button type="button" class="btn btn-outline-primary">Primary</button>
<button type="button" class="btn btn-outline-secondary">Secondary</button>
<button type="button" class="btn btn-outline-success">Success</button>
<button type="button" class="btn btn-outline-info">Info</button>
<button type="button" class="btn btn-outline-warning">Warning</button>
<button type="button" class="btn btn-outline-danger">Danger</button>
<button type="button" class="btn btn-outline-dark">Dark</button>
<button type="button" class="btn btn-outline-light text-dark">Light</button>
```



```
<button type="button" class="btn btn-primary btn-lg">Large</button>
<button type="button" class="btn btn-primary">Default</button>
<button type="button" class="btn btn-primary btn-sm">Small</button>
```



```
<button class="btn btn-primary">
  <span class="spinner-border spinner-border-sm"></span>
</button>
<button class="btn btn-primary">
  <span class="spinner-border spinner-border-sm"></span>
  Loading..
</button>
<button class="btn btn-primary" disabled>
  <span class="spinner-border spinner-border-sm"></span>
  Loading..
</button>
<button class="btn btn-primary" disabled>
  <span class="spinner-grow spinner-grow-sm"></span>
  Loading..
</button>
```



```
<div class="btn-group">
  <button type="button" class="btn btn-primary">Apple</button>
  <button type="button" class="btn btn-primary">Samsung</button>
  <div class="btn-group">
    <button type="button" class="btn btn-primary dropdown-toggle"
data-toggle="dropdown">
      Sony
    </button>
    <div class="dropdown-menu">
      <a class="dropdown-item" href="#">Tablet</a>
      <a class="dropdown-item" href="#">Smartphone</a>
    </div>
  </div>
</div></div>
```

```

<div class="btn-group-vertical">
  <button type="button" class="btn btn-primary">Apple</button>
  <button type="button" class="btn btn-primary">Samsung</button>
  <div class="btn-group">
    <button type="button" class="btn btn-primary dropdown-toggle"
data-toggle="dropdown">
      Sony
    </button>
    <div class="dropdown-menu">
      <a class="dropdown-item" href="#">Tablet</a>
      <a class="dropdown-item" href="#">Smartphone</a>
    </div>
  </div>
</div>

```



```

<div class="spinner-border text-muted"></div>
<div class="spinner-border text-primary"></div>
<div class="spinner-border text-success"></div>
<div class="spinner-border text-info"></div>
<div class="spinner-border text-warning"></div>
<div class="spinner-border text-danger"></div>
<div class="spinner-border text-secondary"></div>
<div class="spinner-border text-dark"></div>
<div class="spinner-border text-light"></div>

```

## Pagination

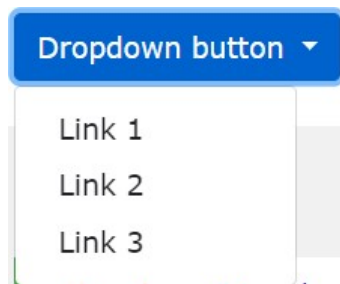
|          |   |   |   |      |
|----------|---|---|---|------|
| Previous | 1 | 2 | 3 | Next |
|----------|---|---|---|------|

```
<ul class="pagination">
  <li class="page-item"><a class="page-link" href="#">Previous</a></li>
  <li class="page-item"><a class="page-link" href="#">1</a></li>
  <li class="page-item active"><a class="page-link" href="#">2</a></li>
  <li class="page-item"><a class="page-link" href="#">3</a></li>
  <li class="page-item"><a class="page-link" href="#">Next</a></li>
</ul>
```

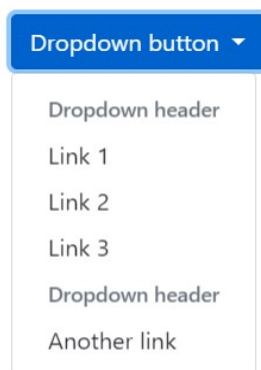
```
<ul class="pagination pagination-lg">
  <li class="page-item"><a class="page-link" href="#">Previous</a></li>
  <li class="page-item"><a class="page-link" href="#">1</a></li>
  <li class="page-item"><a class="page-link" href="#">2</a></li>
  <li class="page-item"><a class="page-link" href="#">3</a></li>
  <li class="page-item"><a class="page-link" href="#">Next</a></li>
</ul>
```

```
<ul class="pagination pagination-sm">
  <li class="page-item"><a class="page-link" href="#">Previous</a></li>
  <li class="page-item"><a class="page-link" href="#">1</a></li>
  <li class="page-item"><a class="page-link" href="#">2</a></li>
  <li class="page-item"><a class="page-link" href="#">3</a></li>
  <li class="page-item"><a class="page-link" href="#">Next</a></li>
</ul>
```

## Dropdown



```
<div class="dropdown">
  <button type="button" class="btn btn-primary dropdown-toggle"
data-toggle="dropdown">
    Dropdown button
  </button>
  <div class="dropdown-menu">
    <a class="dropdown-item" href="#">Link 1</a>
    <a class="dropdown-item" href="#">Link 2</a>
    <a class="dropdown-item" href="#">Link 3</a>
  </div>
</div>
```



```
<div class="container">
  <h2>Dropdowns</h2>
  <p>The .dropdown-header class is used to add headers inside the dropdown
menu:</p>
  <div class="dropdown">
    <button type="button" class="btn btn-primary dropdown-toggle"
data-toggle="dropdown">
      Dropdown button
    </button>
```

```

<div class="dropdown-menu">
  <h5 class="dropdown-header">Dropdown header</h5>
  <a class="dropdown-item" href="#">Link 1</a>
  <a class="dropdown-item" href="#">Link 2</a>
  <a class="dropdown-item" href="#">Link 3</a>
  <h5 class="dropdown-header">Dropdown header</h5>
  <a class="dropdown-item" href="#">Another link</a>
</div>
</div>
</div>

```

## Form

Email:

Password:

☐ Remember me

```

<form action="">
  <div class="form-group">
    <label for="email">Email address:</label>
    <input type="email" class="form-control" placeholder="Enter email"
id="email">
  </div>
  <div class="form-group">
    <label for="pwd">Password:</label>
    <input type="password" class="form-control" placeholder="Enter password"
id="pwd">
  </div>
  <div class="form-group form-check">
    <label class="form-check-label">
      <input class="form-check-input" type="checkbox"> Remember me
    </label>
  </div>

```

```
<button type="submit" class="btn btn-primary">Submit</button>
</form>
```

You can use different validation classes to provide valuable feedback to users. Add either `.was-validated` or `.needs-validation` to the `<form>` element, depending on whether you want to provide validation feedback before or after submitting the form. The input fields will have a **green (valid)** or **red (invalid)** border to indicate what's missing in the form. You can also add a `.valid-feedback` or `.invalid-feedback` message to tell the user explicitly what's missing, or needs to be done before submitting the form.

---

## Form Validation

In this example, we use `.was-validated` to indicate what's missing before submitting the form:

Username:

Please fill out this field.

Password:

Please fill out this field.

☐ I agree on blabla.

Check this checkbox to continue.

```
<form action="" class="was-validated">
  <div class="form-group">
    <label for="uname">Username:</label>
    <input type="text" class="form-control" id="uname" placeholder="Enter
username" name="uname" required>
    <div class="valid-feedback">Valid.</div>
    <div class="invalid-feedback">Please fill out this field.</div>
  </div>
  <div class="form-group">
    <label for="pwd">Password:</label>
    <input type="password" class="form-control" id="pwd" placeholder="Enter
password" name="pswd" required>
    <div class="valid-feedback">Valid.</div>
```

```

    <div class="invalid-feedback">Please fill out this field.</div>
  </div>
  <div class="form-group form-check">
    <label class="form-check-label">
      <input class="form-check-input" type="checkbox" name="remember"
required> I agree on blabla.
    <div class="valid-feedback">Valid.</div>
    <div class="invalid-feedback">Check this checkbox to continue.</div>
  </label>
</div>
<button type="submit" class="btn btn-primary">Submit</button>
</form>

```

In this example, we use `.needs-validation`, which will add the validation effect AFTER the form has been submitting (if there's anything missing). Note that you will also have to add some jQuery code for this example to work properly:

```

<form action="" class="needs-validation" novalidate>
  <div class="form-group">
    <label for="uname">Username:</label>
    <input type="text" class="form-control" id="uname" placeholder="Enter
username" name="uname" required>
    <div class="valid-feedback">Valid.</div>
    <div class="invalid-feedback">Please fill out this field.</div>
  </div>
  <div class="form-group">
    <label for="pwd">Password:</label>
    <input type="password" class="form-control" id="pwd" placeholder="Enter
password" name="pswd" required>
    <div class="valid-feedback">Valid.</div>
    <div class="invalid-feedback">Please fill out this field.</div>
  </div>
  <div class="form-group form-check">
    <label class="form-check-label">
      <input class="form-check-input" type="checkbox" name="remember"
required> I agree on blabla.
    <div class="valid-feedback">Valid.</div>
    <div class="invalid-feedback">Check this checkbox to continue.</div>
  </label>
</div>

```

```
        </label>
    </div>
    <button type="submit" class="btn btn-primary">Submit</button>
</form>

<script>
// Disable form submissions if there are invalid fields
(function() {
    'use strict';
    window.addEventListener('load', function() {
        // Get the forms we want to add validation styles to
        var forms = document.getElementsByClassName('needs-validation');
        // Loop over them and prevent submission
        var validation = Array.prototype.filter.call(forms, function(form) {
            form.addEventListener('submit', function(event) {
                if (form.checkValidity() === false) {
                    event.preventDefault();
                    event.stopPropagation();
                }
                form.classList.add('was-validated');
            }, false);
        });
    }, false);
})();
</script>
```

## Create a Modal

```
<!-- Button to Open the Modal -->
<button type="button" class="btn btn-primary" data-toggle="modal"
data-target="#myModal">
  Open modal
</button>

<!-- The Modal -->
<div class="modal" id="myModal">
  <div class="modal-dialog">
    <div class="modal-content">

      <!-- Modal Header -->
      <div class="modal-header">
        <h4 class="modal-title">Modal Heading</h4>
        <button type="button" class="close"
data-dismiss="modal">&times;</button>
      </div>

      <!-- Modal body -->
      <div class="modal-body">
        Modal body..
      </div>

      <!-- Modal footer -->
      <div class="modal-footer">
        <button type="button" class="btn btn-danger"
data-dismiss="modal">Close</button>
      </div>
    </div>
  </div>
</div>
```

## NavBar

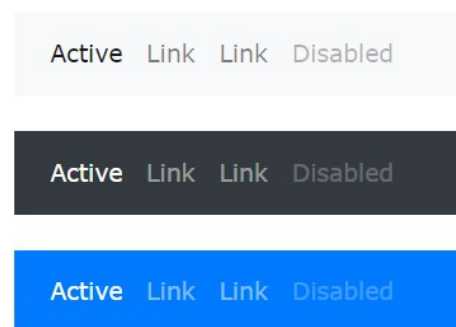
A standard navigation bar is created with the `.navbar` class, followed by a responsive collapsing class: `.navbar-expand-xl|lg|md|sm` (stacks the navbar vertically on extra large, large, medium or small screens).

To add links inside the navbar, use a `<ul>` element with `class="navbar-nav"`. Then add `<li>` elements with a `.nav-item` class followed by an `<a>` element with a `.nav-link` class:

```
<nav class="navbar navbar-expand-sm bg-light">
  <!-- Links -->
  <ul class="navbar-nav">
    <li class="nav-item">
      <a class="nav-link" href="#">Link 1</a>
    </li>
    <li class="nav-item">
      <a class="nav-link" href="#">Link 2</a>
    </li>
    <li class="nav-item">
      <a class="nav-link" href="#">Link 3</a>
    </li>
  </ul>
</nav>
```

## Colored Navbar

Use any of the `.bg-color` classes to change the background color of the navbar (`.bg-primary`, `.bg-success`, `.bg-info`, `.bg-warning`, `.bg-danger`, `.bg-secondary`, `.bg-dark` and `.bg-light`)



```

<!-- Grey with black text -->
<nav class="navbar navbar-expand-sm bg-light navbar-light">
  <ul class="navbar-nav">
    <li class="nav-item active">
      <a class="nav-link" href="#">Active</a>
    </li>
    <li class="nav-item">
      <a class="nav-link" href="#">Link</a>
    </li>
    <li class="nav-item">
      <a class="nav-link" href="#">Link</a>
    </li>
    <li class="nav-item">
      <a class="nav-link disabled" href="#">Disabled</a>
    </li>
  </ul>
</nav>

```

```

<!-- Black with white text -->
<nav class="navbar navbar-expand-sm bg-dark navbar-dark">...</nav>

```

```

<!-- Blue with white text -->
<nav class="navbar navbar-expand-sm bg-primary navbar-dark">...</nav>

```



```

<nav class="navbar navbar-expand-sm bg-dark navbar-dark">
  <form class="form-inline" action="/action_page.php">
    <input class="form-control mr-sm-2" type="text" placeholder="Search">
    <button class="btn btn-success" type="submit">Search</button>
  </form>
</nav>

```



John Doe

Some example text some  
example text. John Doe is an  
architect and engineer

See Profile

```
<div class="card" style="width:400px">
  
  <div class="card-body">
    <h4 class="card-title">John Doe</h4>
    <p class="card-text">Some example text.</p>
    <a href="#" class="btn btn-primary">See Profile</a>
  </div>
</div>
```

Active Dropdown ▾ Link Disabled

```
<ul class="nav nav-pills">
  <li class="nav-item">
    <a class="nav-link active" href="#">Active</a>
  </li>
  <li class="nav-item dropdown">
    <a class="nav-link dropdown-toggle" data-toggle="dropdown"
href="#">Dropdown</a>
    <div class="dropdown-menu">
      <a class="dropdown-item" href="#">Link 1</a>
      <a class="dropdown-item" href="#">Link 2</a>
      <a class="dropdown-item" href="#">Link 3</a>
    </div>
  </li>
```

```
<li class="nav-item">
  <a class="nav-link" href="#">Link</a>
</li>
<li class="nav-item">
  <a class="nav-link disabled" href="#">Disabled</a>
</li>
</ul>
```